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Wax protection for in-canal hearing device

Abstract

A hearing aid assembly includes a receiver speaker enclosed within a receiver housing. The receiver housing defines a circuitous receiver acoustic channel, wherein the receiver channel defines an open side along at least a portion of the receiver channel. The receiver channel also defines a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, wherein the pass-through passage is partially defined by the wall defining a receiver opening. The hearing aid assembly also includes an earbud configured to fit over at least a portion of the receiver housing and partially cover the open side of the receiver channel.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/101,625, filed Jan. 26, 2023, which is a continuation of U.S. patent application Ser. No. 17/237,579, filed Apr. 22, 2021, and issued as U.S. Pat. No. 11,595,769 on Feb. 28, 2023, which is

a continuation of U.S. patent application Ser. No. 16/685,331, filed Nov. 15, 2019, and issued as U.S. Pat. No. 10,993,054 on Apr. 27, 2021, which claims the benefit of U.S. Provisional Application No. 62/770,372, filed Nov. 21, 2018, the content of which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Embodiments herein relate to protecting hearing aid components from damage during use.

BACKGROUND OF THE INVENTION

(2) Hearing aid components are often positioned within a user's ear canal during use, where they provide amplified or otherwise processed sound to the user's ear. A receiver housing is an example of a hearing aid component that can be positioned in a user's ear canal. The receiver housing encloses a receiver speaker, commonly referred to as a receiver, which generates the amplified or processed sound. Earwax accumulation can damage hearing aid components, such as a receiver speaker, if earwax enters the housing of the component. Some hearing aid components have structures that allow earwax to easily enter a receiver housing. Failure to clean a hearing aid component, such as the receiver housing and openings in the receiver housing, to remove accumulated earwax can negatively impact acoustic performance of the device and lead to problems with device function.

(3) Some hearing aid components include replaceable barriers to reduce the likelihood of the earwax entering a receiver housing. The barriers can become clogged and then significantly reduce the sound delivered to the user's ear. Even before becoming clogged, the barriers can reduce acoustic performance of the device. The barriers can be difficult to replace due to their tiny size. Lack of replacement of these barriers can lead to early failure of the hearing aid.

SUMMARY OF THE INVENTION

(4) One general aspect includes a hearing aid assembly including: a receiver assembly including a receiver speaker and a receiver housing defining a receiver cavity to enclose the receiver assembly. The receiver housing defines a circuitous receiver acoustic channel having an open side along at least a portion of the receiver channel. The receiver housing also defines a receiver opening defined in a wall of the receiver channel and in acoustic communication with the receiver cavity. The receiver housing also defines a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, where the pass-through passage is partially defined by the wall defining the receiver opening. The hearing aid assembly also includes an earbud configured to fit over at least a portion of the receiver housing, the earbud including an earbud inner surface that at least partially covers the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing.

(5) Implementations may include one or more of the following features. The receiver channel is tortuous. The receiver channel includes at least three segments, where each the three segments define an angle with an adjacent segment of 90 degrees or less. The receiver channel defines a receiver acoustic path having at least four turns of 90 degrees or less. The receiver channel includes a front segment defined in a front face of the receiver housing, a side segment defined in a side surface of the receiver housing, and a rear segment extending from the side surface of the receiver housing to the receiver opening. The open side of the receiver channel permits a line of sight to a bottom surface of the receiver channel. The open side of the receiver channel has a width of at least 1 millimeter and the receiver channel has a depth of at least 1 millimeter. The receiver channel includes a wrap-around segment that wraps around a circumference of a side surface of the receiver housing. The receiver housing includes a front receiver housing and a back receiver housing connectable to form the receiver cavity, where the front receiver housing defines the receiver channel, the receiver opening, and the pass-through passage. The front receiver housing is a unitary structure. The pass-through passage is perpendicular to an axis of the receiver opening. The receiver opening includes two separate apertures. The receiver opening includes three separate

apertures, where the wall defining the receiver opening includes two ribs, each of the ribs separating two of the three separate apertures. The receiver housing further defines a cavity passage extending from a receiver opening to a cavity opening, where the cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the receiver opening toward the cavity passage. The receiver housing further includes a filter positioned across the receiver opening, where the filter is a wire mesh, a membrane, a foam, or a polystyrene foam. The filter is secured to the receiver housing by over molding, adhesive, or plastic welding. The earbud inner surface seals against the receiver housing over at least a portion of the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing. The earbud includes a flexible material, a resilient material, or an elastomeric material.

(6) One general aspect includes a hearing aid assembly including a receiver assembly including a receiver speaker and a receiver housing defining a receiver cavity to enclose the receiver assembly. The receiver housing defines a circuitous receiver acoustic channel, where at least a portion of the receiver channel defines an open side. The receiver channel includes a front segment defined in a front face of the receiver housing, a side segment defined in a side surface of the receiver housing, and a rear segment extending from the side surface toward a longitudinal axis of the receiver housing and to the receiver opening, where each the front, side, and rear segments define an angle with an adjacent segment of 90 degrees or less. The hearing aid assembly also includes a receiver opening defined in a wall of the receiver channel in acoustic communication with the receiver cavity, where a cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the receiver opening to the receiver cavity. The hearing aid assembly also includes a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, where the pass-through passage includes the rear segment of the receiver channel, where the pass-through passage is transverse to a longitudinal axis of the receiver opening. The hearing aid assembly also includes an earbud sized to fit over at least a portion of the receiver housing, the earbud including an earbud inner surface that at least partially covers the open side of the receiver channel and seals against the receiver housing over at least a portion of the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing.

(7) One general aspect includes a method of making a hearing aid assembly including providing a receiver assembly including a receiver speaker and providing a receiver housing defining a receiver cavity to enclose the receiver assembly. The receiver housing defines a circuitous receiver acoustic channel, where the receiver channel defines an open side along at least a portion of the receiver channel, a receiver opening defined in a wall of the receiver channel and configured to be in acoustic communication with the receiver speaker, and a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, where the pass-through passage is partially defined by the wall defining the receiver opening. The method also includes assembling the receiver housing around the receiver assembly. The method also includes providing an earbud sized to fit over at least a portion of the receiver housing, the earbud including an earbud inner surface that at least partially covers the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing.

(8) This summary is an overview of some of the teachings of the present application and is not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details are found in the detailed description and appended claims. Other aspects will be apparent to persons skilled in the art upon reading and understanding the following detailed description and viewing the drawings that form a part thereof, each of which is not to be taken in a limiting sense. The scope herein is defined by the appended claims and their legal equivalents.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Aspects may be more completely understood in connection with the following figures (FIGS.), in which:

(2) FIG. 1 is a perspective view of a hearing aid system including a behind-the-ear unit, a cable, a receiver housing, and an earbud in accordance with various embodiments herein.

(3) FIG. 2 is a perspective view of the receiver housing and earbud of FIG. 1 in accordance with various embodiments herein.

(4) FIG. 3 is a perspective, exploded view of the receiver housing and earbud of FIG. 1 in accordance with various embodiments herein.

(5) FIG. 4 is a cross-sectional view of the receiver housing and earbud of FIG. 1, in accordance with various embodiments herein.

(6) FIG. 5 is a perspective view of the receiver housing of FIG. 1 in accordance with various embodiments herein.

(7) FIG. 6 is a cross-sectional perspective view of a front receiver housing of the receiver housing of FIG. 1 in accordance with various embodiments herein.

(8) FIG. 7 is a cross-sectional side view of the front receiver housing of FIG. 6 in accordance with various embodiments herein.

(9) FIGS. 8 and 9 are perspective views from different angles of an alternate front receiver housing in accordance with various embodiments herein.

(10) FIG. 10 is a side view of the front receiver housing of FIG. 8.

(11) FIG. 11 is a cross-sectional view of the front receiver housing of FIG. 8 along line 11-11 in FIG. 10.

(12) FIG. 12 is a cross-sectional view of the front receiver housing of FIG. 8 along line 12-12 of FIG. 11.

(13) FIG. 13 is a cross-sectional view of an alternate front receiver housing from the same cross-sectional viewpoint as FIG. 12, including a filter in the receiver opening.

(14) FIG. 14 is a perspective side view of an alternate front receiver housing having a wrap-around segment of a circuitous receiver acoustic channel.

(15) FIG. 15 is a perspective view of a cleaning tool for use with a hearing aid system described herein.

(16) FIG. 16 is an alternate receiver housing according to various examples.

(17) FIG. 17 is a side perspective view of a receiver housing having an alternate retaining stem in accordance with various embodiments herein.

(18) FIG. 18 is another side perspective view of the receiver housing of FIG. 17 in accordance with various embodiments herein.

(19) FIG. 19 is a top end perspective view of the receiver housing of FIG. 17 in accordance with various embodiments herein.

(20) FIG. 20 is a cross-sectional view of the receiver housing of FIG. 17 in accordance with various embodiments herein.

(21) While embodiments are susceptible to various modifications and alternative forms, specifics thereof have been shown by way of example and drawings and will be described in detail. It should be understood, however, that the scope herein is not limited to the particular aspects described. On the contrary, the intention is to cover modifications, equivalents, and alternatives falling within the spirit and scope herein.

DETAILED DESCRIPTION

(22) A hearing aid assembly has a receiver speaker enclosed within a receiver housing, where the receiver housing defines a receiver opening in acoustic communication with the receiver speaker. To reduce earwax entry into the receiver housing, the receiver housing defines a circuitous receiver acoustic channel leading from a front end of the receiver housing to the receiver opening. Earwax

must fill the circuitous path and navigate multiple turns in order to fully block the acoustic path and reach the receiver opening. The circuitous receiver acoustic channel will be referred to as the receiver channel herein, for brevity, at times. An earbud that fits over the receiver housing provides the final wall over an open side of the receiver channel. The receiver housing also creates a shelf which blocks direct earwax entry into the receiver opening. This shelf is open on both ends, defining a pass-through passage through the receiver housing. The pass-through passage facilitates easier cleaning of the receiver channel. The open side of the receiver channel also facilitates easier cleaning.

(23) In some examples, the receiver channel includes at least three segments, and each segment defines an angle with an adjacent segment of 90 degrees or less. In some examples, the receiver channel includes at least four segments, and each segment defines an angle with an adjacent segment of 90 degrees or less.

(24) In some examples, the pass-through passage is perpendicular to an axis of the receiver opening. A cleaning motion with a cleaning tool through the pass-through passage will therefore be perpendicular to the receiver opening, and less likely to push earwax into the receiver opening. In some examples, a filter is integrated into the receiver opening and positioned over the receiver opening to reduce earwax and debris from entering the receiver housing through the receiver opening.

(25) The receiver opening in some examples includes multiple apertures. In some examples, ribs divide the receiver opening into multiple apertures. The ribs can serve as an additional barrier to earwax to prevent it or increase the time before it reaches the inside of the receiver housing. The receiver opening in some embodiments is tapered, so that it is narrower closer to the receiver channel and wider closer to the receiver assembly. In addition, or alternatively, if there are multiple apertures making up the receiver opening, the apertures can be tapered to be narrower near the receiver channel and wider closer to the receiver assembly. This configuration can reduce the likelihood of liquid entering the receiver housing through the receiver opening if the receiver housing comes into contact with liquid, such as if liquid is used to clean the receiver housing or the receiver housing accidentally contacts liquid.

(26) Examples of hearing aid systems and receiver housings, where the receiver housing has a circuitous receiver acoustic channel and defines a pass-through passage, are shown in the FIGS. and will now be described. FIG. 1 is a perspective view of a hearing aid system **100** including an external unit **101**, a receiver housing **102**, an earbud **104** covering a portion of the receiver housing **102**, and a cable **106** connecting the external unit **101** with the receiver housing **102**, in accordance with various embodiments herein. The external unit **101** may be worn outside of the ear canal, such as over the user's ear, behind the user's ear, clipped to a user's clothing, or many other locations.

(27) A “hearing aid system” as used herein can include many different devices and combinations of devices that aid a person with impaired hearing by producing amplified sound. A “hearing aid system” can also refer to devices that produce optimized or processed sound for a user with normal hearing. The amplified, optimized or processed sound is output to a user, such as into the ear canal of a user.

(28) Components of a hearing aid system device herein can include a control circuit, digital signal processor (DSP), memory (such as non-volatile memory), power management circuitry, a data communications bus, one or more communication devices (e.g., a radio, a near-field magnetic induction device), one or more antennas, one or more microphones, and various sensors as described in greater detail below. More advanced hearing assistance devices can incorporate a long-range communication device, such as a Bluetooth® transceiver or other type of radio frequency (RF) transceiver. The hearing aid system can define a battery compartment into which a battery can be disposed to provide power to the device. These components can be divided between the external unit **101**, other external devices, and the receiver housing **102**. An external unit **101** can include input devices such as buttons or pads to control the hearing aid system. The receiver housing **102**

encloses a speaker, also referred to as a receiver speaker or a receiver. The receiver housing **102** is sized and shaped to fit within a user's ear canal. The hearing aid system **100** can be referred to as a receiver-in-canal (RIC) system. Sound is output from the receiver housing **102** to the user's ear canal. The cable **106** can include one or more electrical conductors and provide electrical communication between components inside of the external unit **101** and components inside of the receiver housing **102**. The cable **106** provides an electrical signal from the external unit to drive the receiver to produce sound.

(29) FIG. **2** is a perspective view of the receiver housing and earbud of FIG. **1** in accordance with various embodiments herein. Sound from the receiver within the receiver housing **102** travels an acoustic path through the receiver housing **102** and earbud **104** and exits at a front earbud opening **202** of the earbud **104**. The earbud **104** includes a wax bridge **204** over the front earbud opening **202** and an earbud outer surface **206**.

(30) FIG. **3** is a perspective, exploded view of the receiver housing **102** and earbud **104** of FIG. **1** in accordance with various embodiments herein. The receiver housing **102** defines a cavity **302** in which the receiver assembly **304** is enclosed. In one example, the receiver housing **102** is made up of a front receiver housing **306** joined to a rear receiver housing **308**. The receiver assembly **304** includes the receiver speaker or receiver that converts electrical impulses into sound, such as an electroacoustic transducer, speaker, or loud speaker.

(31) The front receiver housing **306** defines a circuitous receiver acoustic channel **310**, and a portion of it is visible in FIG. **3** at a front end **312** of the front receiver housing **306**. The circuitous receiver acoustic channel **310**, also referred to as the receiver channel **310**, forms part of an acoustic path traveled by sound from the receiver assembly **304** to the front earbud opening **202** and into the user's ear canal.

(32) The front receiver housing **306** extends between the front end **312** and the rear end **314** and includes a side surface **316**. The rear end **314** of the front receiver housing **306** is connected to the rear receiver housing **308** to form the receiver cavity **302** around the receiver assembly **304**. The receiver channel **310** terminates at a front end **318** defined by the front end **312** of the front receiver housing.

(33) An acoustic path **402** defined by of the receiver housing and earbud will now be described with respect to FIG. **4**. FIG. **4** is a cross-sectional view of the receiver housing and earbud of FIG. **1**, in accordance with various embodiments herein. Upon receiving electrical signals via the cable, sound is generated by a receiver **404** within the cavity **302** of the receiver housing **102**. The sound travels the acoustic path **402** through a receiver opening **406**. The receiver opening **406** is defined in a wall of the receiver channel **310** and is configured to be in acoustic communication with the receiver **404**. After leaving the receiver opening **406**, the sound travels along multiple segments of the circuitous receiver acoustic channel **310**, through an earbud acoustic channel **408**, and out the front earbud opening **202** to reach the user's ear canal.

(34) The circuitous receiver acoustic channel **310**, or receiver channel **310**, is defined by the structure of the front receiver housing **306** and extends from the receiver opening **406** to a front end of the receiver channel **310** defined by the front end **318** of the front receiver housing **306**. The receiver channel **310** is not a direct path between the receiver opening **406** and the front end **312** of the front receiver housing **306**. The receiver channel **310** travels an indirect, circuitous path. In some examples, the receiver channel **310** is tortuous. By "tortuous", it is meant that the receiver channel **310** is twisting and serpentine. In various examples, the receiver channel includes at least two segments, at least three segments, and at least four segments. In various examples, each segment of the receiver channel **310** define an angle with an adjacent segment of 90 degrees or less. In various examples, the receiver channel has at least four turns of 90 degrees or less.

(35) In the example of FIGS. **4-12**, the receiver channel **310** defined in the front receiver housing **306** includes a front segment **412** defined in the front end **312**, a side segment **414** defined in the side surface **316**, and a rear segment **416** extending from the side surface **316** to the receiver

opening **406**. These segments are at 90-degree angles to each other.

(36) In various embodiments, a portion of the receiver channel **310** has an open side that is not bounded by the structure of front receiver housing **306**. In various embodiments, the receiver channel **310** is structured to permit a line of sight to a bottom surface of the receiver channel **310**. When the earbud **104** is placed over a portion of the front receiver housing **306**, an earbud inner surface **420** closes the open side of the receiver channel over that portion. When the earbud **104** is removed, the open channel portion of the receiver channel **310** is easy to clean because the user has a line of sight into the receiver channel **310**.

(37) The earbud **104** includes an axial wall **422** that surrounds a portion of the front receiver housing **306** when the earbud **104** is placed over a portion of the front receiver housing **306**. The axial wall **422** defines and surrounds the earbud acoustic channel **408**. In various embodiments, the earbud **104** also includes an outer dome **424**. In various embodiments, the earbud **104** is made from a material and constructed so that it uniformly conforms to the ear canal and maintains a constant and comfortable radial pressure on the ear canal. In various examples, the earbud **104** is made of resilient material, such as silicone. In various examples, the earbud **104** is made of a flexible material. By flexible material, it is meant that a material is capable of bending easily without breaking. In various examples, the earbud is made of an elastomeric material. By “elastomeric material”, it is meant a material with viscoelasticity that is soft and deformable at ambient temperatures, such as rubber, silicone, and amorphous polymers. The flexibility and resilience of the material facilitates a seal of the earbud inner surface **420** to the front receiver housing **306** over the receiver channel **310**. Examples of an earbud **104** are shown in commonly assigned U.S. Pat. No. 9,479,878, titled, “Enhanced Comfort Earbud,” issued Oct. 25, 2016, which is incorporated by reference herein in its entirety. Many other configurations of earbud can be used with the hearing aid system described herein.

(38) Pass-Through Passage

(39) The receiver channel **310** and the acoustic path **402** partially defined by the receiver channel **310** overlaps with a pass-through passage **430** that is defined in the front receiver housing **306** and extends from a first side to a second side of the receiver housing **102**. The pass-through passage **430** is bounded on one side by the wall or shelf **432** that defines the receiver opening **406**. The shelf **432** is open on both ends. The pass-through passage facilitates easier cleaning of the receiver channel.

(40) Another view of the pass-through passage **430** and the receiver channel **310** is provided in FIG. 5, which is a perspective view of the receiver housing of FIG. 1 in accordance with various embodiments herein. The wall or shelf **432** defines the receiver opening **406**. The receiver opening **406** includes multiple circular apertures.

(41) The open side of the receiver channel **310** is visible in FIG. 5. The receiver channel **310** has an open side along the front segment **412**, side segment **414**, and rear segment **416**. In various embodiments, the width of the open side of the receiver channel **310** is at least about 1 millimeter. In various embodiments, the depth of the receiver channel **310** is at least about 1 millimeter. In various embodiments, the open channel is shaped so that a user has visibility to the entire receiver channel surface.

(42) FIG. 6 is a cross-sectional perspective view and FIG. 7 is a cross-sectional side view of a front receiver housing **306** in accordance with various embodiments herein, separated from the rear receiver housing (not shown). The cross-sectional plane of FIGS. 6 and 7 slices through the open side **602** of the receiver channel **310**. The receiver opening **406** shown in FIGS. 6 and 7 is different than the receiver opening **406** of FIG. 5 and includes a rib **604** that divides the receiver opening into multiple apertures. In various embodiments, one rib, two ribs, three ribs or another number of ribs are provided that extend across the receiver opening **406** and create two, three, four, or another number of multiple apertures.

(43) FIG. 7 shows a longitudinal axis L-L of the front receiver housing, which is also the

longitudinal axis of the receiver opening **406**. FIG. 7 also shows a longitudinal axis P-P of the pass-through passage **430**. In various embodiments, the longitudinal axis of the receiver opening **406** is transverse to the longitudinal axis of the pass-through passage **430**. A cleaning motion with a cleaning tool through that portion of the receiver channel **310** will therefore be transverse to the receiver opening, and less likely to push earwax into the receiver opening than if a cleaning motion was in the same direction as the axis of the receiver opening. In various embodiments, such as shown in FIGS. 6 and 7, the longitudinal axis of the receiver opening **406** is approximately perpendicular to the longitudinal axis of the pass-through passage **430**. In various embodiments, the longitudinal axis of the front receiver housing **306** is transverse to the longitudinal axis of the pass-through passage. In various embodiments, such as shown in FIGS. 6 and 7, the longitudinal axis of the front receiver housing **306** is approximately perpendicular to the longitudinal axis of the pass-through passage **430**. The receiver channel depth D shown in FIG. 7 is at least 1 millimeter in various embodiments.

(44) As seen in FIG. 7, a cavity passage **702** extends from the receiver opening **406** to a cavity opening **704** adjacent to the cavity **302**. In various embodiments, the receiver opening **406** has a smaller largest dimension, a smaller cross-sectional area, or both than the cavity opening **704**. The cavity passage **702** tapers to a smaller dimension at the receiver opening **406**. In other words, the cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the pass-through passage. This configuration can reduce the likelihood of liquid entering the receiver housing through the receiver opening if the receiver housing comes into contact with liquid, such as if liquid is used to clean the receiver housing or the receiver housing accidentally contacts liquid.

(45) Front Receiver Housing of FIGS. 8-12

(46) An alternate embodiment of a front receiver housing **306** is shown from different viewpoints in FIGS. 8-12. Like reference numbers refer to similar parts in FIGS. 8-12 compared to FIGS. 1-7. Like the embodiments of FIGS. 1-7, the front receiver housing **306** of FIGS. 8-12 includes a circuitous acoustic receiver channel **310**, pass-through passage **430**, and defines half of a cavity **302** for the receiver. In various embodiments, the profile of the pass-through passage **430** is D-shaped and is defined by a planar surface formed by the shelf **432** and an arched surface **1102** is shown in FIG. 11.

(47) The receiver opening **406** and cavity passage **1202** of FIGS. 8-12 are different compared to the receiver opening **406** and cavity passage **702** of FIGS. 1-7 and will now be described with reference to FIGS. 11 and 12. FIG. 11 is a cross-sectional view of the front receiver housing of FIG. 8 along line 11-11 in FIG. 10, and FIG. 12 is a cross-sectional view of the front receiver housing of FIG. 8 along line 12-12 of FIG. 11.

(48) The cavity passage **1202** extends from the receiver opening **406** to a cavity opening **1204**. In various embodiments, the receiver opening **406** has a smaller largest dimension, a smaller cross-sectional area, or both than the cavity opening **1204**. The cavity passage **1202** tapers to the smaller dimensions of the receiver opening **406**. In other words, the cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the pass-through passage. This configuration can reduce the likelihood of liquid entering the receiver housing through the receiver opening **406**.

(49) The receiver opening **406** is defined by the shelf **432**, which can include multiple ribs **1206** in various embodiments. The ribs **1206** in the receiver opening **406** provide a barrier for entry of earwax. In various embodiments, two ribs **1206** define three apertures **1208**. In various embodiments, one rib, two ribs, three ribs or another number of ribs are provided that extend across the receiver opening **406** and create two, three, four, or another number of multiple apertures.

(50) The ribs **1206** can have a profile so that the apertures are smaller closer to the pass-through passage **430** than to the cavity **302**. This configuration further reduces the likelihood of liquid entering the cavity **302**, while still allowing sound to travel efficiently to the receiver channel **310**

from the cavity **302**.

(51) Filter

(52) The receiver housing **102** can incorporate a filter into the receiver opening **406**. FIG. **13** is a cross-sectional view of an alternate front receiver housing **306** from the same cross-sectional viewpoint as FIG. **12**, including a filter **1302** across the receiver opening. The filter can be a wire mesh, a membrane, a foam, or a polystyrene foam. The filter can be secured to the receiver housing by over molding, adhesive, or plastic welding.

(53) Wrap-Around Segment

(54) The receiver channel **310** of the receiver housing **102** can take on many different shapes. FIG. **14** is a perspective side view of an alternate front receiver housing **306** with a different receiver channel **310** configuration compared to FIGS. **3-13**. The receiver channel **310** of FIG. **14** extends from the front end **312** of the front receiver housing **306** to the receiver opening (not shown in FIG. **14**). The receiver channel **310** of FIG. **14** includes a front segment **412**, a side segment **414**, and a wrap-around segment **1402** wrapping around a circumference of the side surface **316** of the front receiver housing. The front receiver housing **306** of FIG. **14** also includes a pass-through passage **430** extending from the wrap-around segment **1402** to the receiver opening, though only a small portion of the pass-through passage **430** is shown and the receiver opening is not shown in FIG. **14**.

(55) Receiver Housing

(56) Now referring to FIGS. **3-5**, the receiver housing **102** of the examples described herein include a front receiver housing **306** and a rear receiver housing **308**, extending from a front end or face **312** of the front receiver housing **306** to a rear face **324** of the rear receiver housing **308**. The receiver housing **102** is positioned in the ear canal so that the front end or front face **312** is inserted into the ear first and is positioned more deeply within the ear canal.

(57) The front receiver housing can be a unitary structure in various embodiments, so that the unitary structure cannot be broken down into smaller pieces without destroying the front receiver housing. In various examples, the front receiver housing is a unitary molded structure. The rear receiver housing **308** can also be a unitary structure. The rear receiver housing can be a unitary molded structure.

(58) Instead of a front receiver housing **306** and a rear receiver housing **308**, it is possible for the receiver housing **102** to be divided into two pieces around the cavity **302** in a different configuration, such as two side housings or a receiver housing with a hatch for inserting the receiver assembly **304**. Wherever the front receiver housing **306** is described as defining the receiver channel, defining the pass-through passage, or having other features herein, it should be understood that those descriptions apply to the receiver housing **102** as a whole, however the receiver housing **102** might be divided to give access to the cavity **302**. As a result, the term “receiver housing” could be used accurately in place of “front receiver housing” in most locations herein.

(59) In various examples, the components of the receiver housing **102** are connected to form the cavity **302** for the receiver assembly **304** by using adhesive or plastic bonding. Now referring to FIGS. **2-3**, the receiver housing **102** defines an opening **210** that can be used to attach an accessory device, such as a device that improves retention of the receiver housing in the ear canal. Now referring to FIG. **4**, the receiver housing **102**, and more specifically the rear receiver housing **308** defines a cable inlet **440** where the cable **106** (FIG. **1**) enters the receiver housing **102**.

(60) Alternate Receiver Housing of FIG. **16**

(61) FIG. **16** illustrates an alternative receiver housing **1600** compared to the other FIGS. In the example of FIG. **16**, the receiver housing **1600** includes a behind-the-ear (BTE) receiver housing **1602**, a length of audio tubing **1604**, and an in-canal housing **1606**. The BTE receiver housing **1602** can make up a portion of a larger external unit that is positionable behind the user's ear, similar to external unit **101** of FIG. **1**.

(62) The audio tubing **1604** provides acoustic communication between the BTE receiver housing

1602 and the in-canal housing **1606**.

(63) The in-canal housing **1606** is the portion of the alternative receiver housing **1600** that defines a circuitous receiver acoustic channel **1608** as described herein. The receiver channel **1608** can have any of the configurations described herein. The in-canal housing **1606** can be covered by an ear bud such as the ear bud shown in FIGS. **1-4**. Features of the receiver housings described herein can also be present in the receiver housing of FIG. **16**, including defining an opening in acoustic communication with the receiver speaker in a wall of the receiver channel **1608**. A flexible protrusion **1610** may also be provided to assist with retaining the in-canal housing **1606** in a desired position.

(64) Retaining Stem Configurations

(65) The front receiver housing **306** includes a retaining stem that interfaces with an interior surface of an earbud. The embodiments of FIGS. **1-7** and FIGS. **8-12** have a front receiver housing **306** that includes a retaining stem **502**, best illustrated in FIG. **5** and in FIG. **8**. The retaining stem **502** extends away from a remainder of the front receiver housing **306** and includes a neck portion **504** and a retaining ridge **506**. The retaining ridge **506** is wider from a side perspective than the neck portion **504**, and the neck portion is narrower from a side perspective from the remainder of the front receiver housing **306**. The retaining stem **502** of FIGS. **1-7** and **8-12** has a generally oval shape when viewed from a top perspective, as best seen in FIG. **8**.

(66) A front receiver housing **306** having an alternate embodiment of a retaining stem **1702** is shown from different viewpoints in FIGS. **17-20**. Like reference numbers refer to similar parts in FIGS. **17-20** compared to FIGS. **1-12**. Like the embodiments of FIGS. **1-12**, the retaining stem **1702** is configured to interface with an interior surface of an earbud and extends away from a remainder of the front receiver housing **306**. Like the embodiments of FIGS. **1-12**, the front receiver housing **306** of FIGS. **17-20** includes a circuitous acoustic receiver channel **310**, pass-through passage **430**, and defines half of a cavity **302** for the receiver.

(67) The retaining stem **1702** of the embodiment of FIGS. **17-20** has a different configuration than the retaining stem **502** of FIGS. **1-12**. The retaining stem **1702** includes a neck portion **1704** that is narrower from a side perspective than the remainder of the front receiver housing **306**. In some embodiments, the retaining stem **1702** includes a first radial ridge **1706** near a top end of the retaining stem **1702**. In addition, or in the alternative, the retaining stem **1702** includes a second radial ridge **1708** spaced away from the top end of the retaining stem **1702**. The retaining stem **1702** of FIGS. **17-20** has a generally circular shape when viewed from a top perspective, as best seen in FIG. **19**. The first radial ridge **1706** and the second radial ridge **1708** are both interrupted by the receiver channel **310**. In some embodiments, as shown in FIG. **18** and FIG. **20**, the second radial ridge is interrupted over an opening to the pass-through passage **430**.

(68) Cleaning Tool

(69) A cleaning tool can be used with a hearing aid system as described herein to remove accumulated earwax from a receiver channel, thereby maintaining the audio performance of the system and reducing the likelihood of damage to the components in the receiver channel. The cleaning tool can have an extension that is sized to enter and exit the pass-through passage, can have a portion for scraping wax from the circuitous receiver acoustic channel defined by the receiver housing, and can have a portion with bristles or a cleaning surface that contacts the walls of the pass-through passage. The cleaning tool can be used when the earbud is not covering the receiver channel. The receiver channel is sized so that the user has a line of sight into the receiver channel to monitor the results of cleaning efforts during the cleaning process.

(70) FIG. **15** is a perspective view of one example of a cleaning tool for use with a hearing aid system described herein. The cleaning tool **1500** can include a handle portion **1501** where the user grasps the cleaning tool. The cleaning tool can include a push-through extension **1502** which is designed to enter and exit the pass-through passage and clear wax from the pass-through passage. In one example, the push-through extension **1502** is a flexible monofilament line extending from

one end of the cleaning tool **1500**. In one example, the push-through extension has a length of about 1 inch (2.54 cm). The cleaning tool **1500** can have a scraping tool extension **1504** for scraping wax from the receiver channel. In one example, the scraping tool extension **1504** is a wire loop. The cleaning tool **1500** can also have a brush **1506** sized to brush wax from the receiver channel. It is also possible for the cleaning tool **1500** to include a magnet used to control the hearing aid assembly.

(71) Methods of Making

(72) A method of making a hearing aid assembly includes providing a receiver assembly including a receiver speaker and providing components of a receiver housing that can be assembled to enclose the receiver assembly. The receiver housing defines a circuitous receiver acoustic channel, also referred to as a receiver channel, which has an open side along at least a portion of the receiver channel. The receiver housing also defines a receiver opening in a wall of the receiver channel, where the receiver opening is in acoustic communication with the receiver speaker when the receiver housing is assembled to enclose the receiver speaker. The receiver housing also defines a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, wherein the pass-through passage is partially defined by the wall defining the receiver opening.

(73) The method includes the step of assembling the receiver housing around the receiver assembly. The receiver assembly is then enclosed in a cavity of the receiver housing. An earbud is provided that can fit over at least a portion of the receiver housing. The earbud includes an earbud inner surface that at least partially covers the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing.

(74) Methods of Using

(75) In a method of using the hearing aid assembly, a user obtains a hearing aid assembly including an external unit, a receiver housing connected to the external unit by a cable, and an earbud. The user places the earbud over a portion of the receiver housing that defines the receiver channel. The portion of the receiver housing covered by the earbud includes a front end of the receiver housing. The user secures the external unit in a location outside of the ear canal, such as behind the ear, over the ear, or clipped to clothing. The user inserts the receiver housing and earbud into one of the user's ear canals. The external unit is powered on. The external unit uses a microphone as an input of sound near the user. The external unit provides an electrical signal using the cable to the receiver assembly, causing the receiver assembly to generate amplified, optimized, or processed sound to the user's ear canal. The sound travels from the receiver assembly, out the receiver opening, along the receiver channel which includes a portion of the pass-through passage, through the earbud acoustic path, and out of the front earbud opening into the user's ear canal.

(76) Periodically, the user cleans the receiver housing by removing the earbud from the receiver housing and removing earwax or other debris from the earbud acoustic path and from the receiver channel. The user may use a cleaning tool to brush or push debris from the receiver channel. The open side of the receiver channel allows the user to see the results of the cleaning efforts and see when the accumulated earwax has been removed. The user may push the cleaning tool through the pass-through passage to remove earwax.

(77) It should be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(78) It should also be noted that, as used in this specification and the appended claims, the phrase “configured” describes a system, apparatus, or other structure that is constructed or configured to perform a particular task or adopt a particular configuration. The phrase “configured” can be used interchangeably with other similar phrases such as arranged and configured, constructed and arranged, constructed, manufactured and arranged, and the like.

(79) All publications and patent applications in this specification are indicative of the level of ordinary skill in the art to which this invention pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated by reference.

(80) The embodiments described herein are not intended to be exhaustive or to limit the invention to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art can appreciate and understand the principles and practices. As such, aspects have been described with reference to various specific and preferred embodiments and techniques. However, it should be understood that many variations and modifications may be made while remaining within the spirit and scope herein.

Claims

1. An ear-worn component comprising: a receiver housing defining a receiver cavity to enclose a receiver speaker, the receiver housing defining: a receiver acoustic channel, wherein the receiver channel defines an open side along at least a portion of the receiver channel; a receiver opening defined in a wall of the receiver channel and in acoustic communication with the receiver cavity; and a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, wherein the pass-through passage is partially defined by the wall defining the receiver opening, wherein the pass-through passage is continuous between the first side of the receiver housing and the second side of the receiver housing; and an earbud configured to fit over at least a portion of the receiver housing, the earbud comprising an earbud inner surface that at least partially covers the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing.
2. The ear-worn component of claim 1 wherein the receiver channel is circuitous.
3. The ear-worn component of claim 1 wherein the receiver channel comprises at least three segments, wherein each the three segments define an angle with an adjacent segment of 90 degrees or less.
4. The ear-worn component of claim 1 wherein the receiver channel defines a receiver acoustic path having at least four turns of 90 degrees or less.
5. The ear-worn component of claim 1 wherein the receiver channel comprises: a front segment defined in a front face of the receiver housing, a side segment defined in a side surface of the receiver housing, and a rear segment extending from the side surface of the receiver housing to the receiver opening, wherein the pass-through passage comprises the rear segment of the receiver channel.
6. The ear-worn component of claim 1, wherein the pass-through passage is defined along a pass-through axis extending between the first side of the receiver housing and the second side of the receiver housing.
7. The ear-worn component of claim 1 wherein a tool is configured to enter the pass-through passage at the first side, travel through the pass-through passage, and exit the pass-through passage at the second side.
8. The ear-worn component of claim 1 wherein the receiver channel comprises a wrap-around segment that wraps around a circumference of a side surface of the receiver housing.
9. The ear-worn component of claim 1 wherein the receiver housing comprising a front receiver housing and a back receiver housing connectable to form the receiver cavity, wherein the front receiver housing defines the receiver channel, the receiver opening, and the pass-through passage.
10. The ear-worn component of claim 9 wherein the front receiver housing is a unitary structure.
11. The ear-worn component of claim 1 wherein the pass-through passage is perpendicular to an axis of the receiver opening.
12. The ear-worn component of claim 1 wherein the receiver opening comprises two separate

apertures.

13. The ear-worn component of claim 1 wherein the receiver opening comprises three separate apertures, wherein the wall defining the receiver opening comprises two ribs, each of the ribs separating two of the three separate apertures.

14. The ear-worn component of claim 1 wherein the receiver housing further defines a cavity passage extending from a receiver opening to a cavity opening, wherein the cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the receiver opening toward the cavity passage.

15. The ear-worn component of claim 1 wherein receiver housing further comprises a filter positioned across the receiver opening, wherein the filter is a wire mesh, a membrane, a foam, or a polystyrene foam.

16. The ear-worn component of claim 15 wherein the filter is secured to the receiver housing by over molding, adhesive, or plastic welding.

17. The ear-worn component of claim 1 wherein the earbud inner surface seals against the receiver housing over at least a portion of the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing.

18. The ear-worn component of claim 1 wherein the earbud comprises a flexible material, a resilient material, or an elastomeric material.

19. An ear-worn component comprising: a receiver assembly comprising a receiver speaker; a receiver housing defining a receiver cavity to enclose the receiver assembly, the receiver housing defining: a receiver acoustic channel wherein the receiver channel defines an open side along at least a portion of the receiver channel; receiver opening defined in a wall of the receiver channel in acoustic communication with the receiver cavity, wherein a cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the receiver opening to the receiver cavity; and a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, wherein the pass-through passage is transverse to a longitudinal axis of the receiver opening, wherein the pass-through passage is continuous between the first side of the receiver housing and the second side of the receiver housing; and an earbud configured to fit over at least a portion of the receiver housing, the earbud comprising an earbud inner surface that at least partially covers the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing; and wherein the receiver housing further defines a cavity passage extending from a receiver opening to a cavity opening, wherein the cavity passage is tapered to have an increasing cross-sectional area as the cavity passage extends away from the receiver opening toward the cavity opening.

20. A method of making an ear-worn component comprising: providing a receiver housing defining a receiver cavity to enclose a receiver speaker defining: a receiver acoustic channel wherein the receiver channel defines an open side along at least a portion of the receiver channel; a receiver opening defined in a wall of the receiver channel and configured to be in acoustic communication with the receiver speaker; a pass-through passage extending from a first side of the receiver housing to a second side of the receiver housing, wherein the pass-through passage is partially defined by the wall defining the receiver opening, wherein the pass-through passage is continuous between the first side of the receiver housing and the second side of the receiver housing; and an earbud configured to fit over at least a portion of the receiver housing, the earbud comprising an earbud inner surface that at least partially covers the open side of the receiver channel when the earbud is positioned over at least a portion of the receiver housing; and assembling the receiver housing around the receiver speaker.
